UML Class Diagram

Presented by

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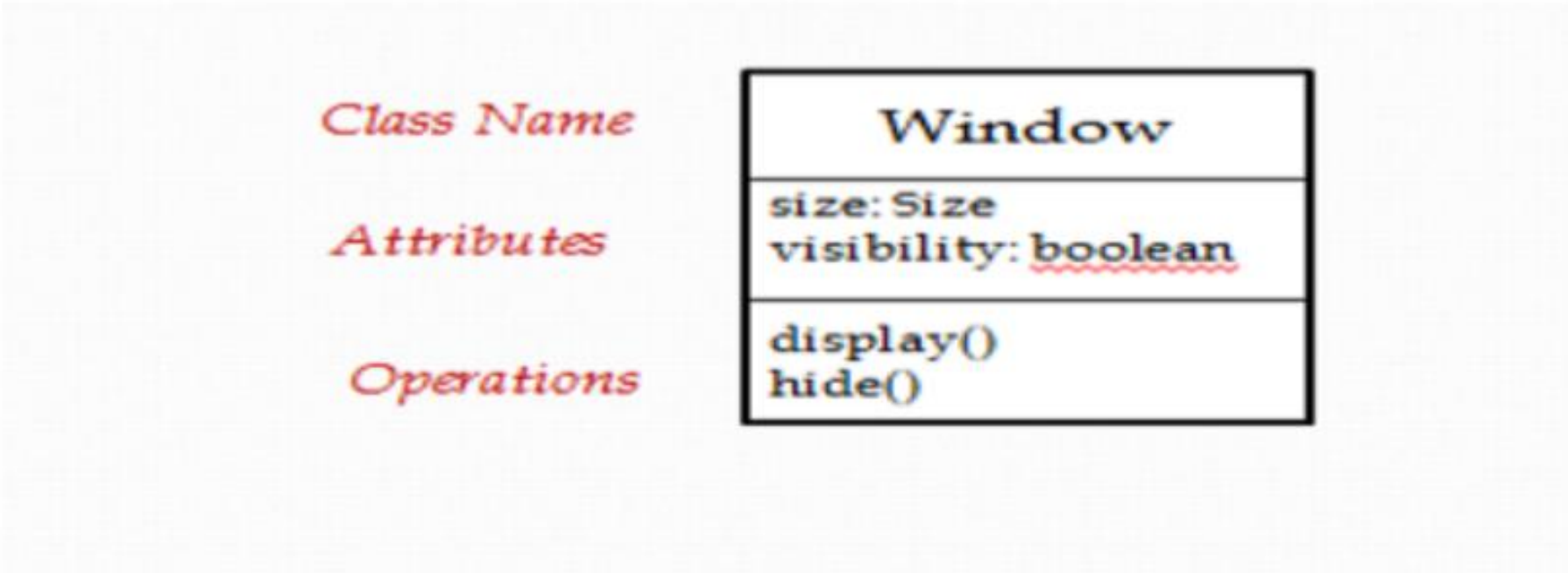


01
Elements of a UML Class Diagram

Classes

Definition of a Class

Describes a set of objects with similar attributes, operations, relationships, and behavior.



Class Components

Class Components

1- **Class Name:** Identifies the class.

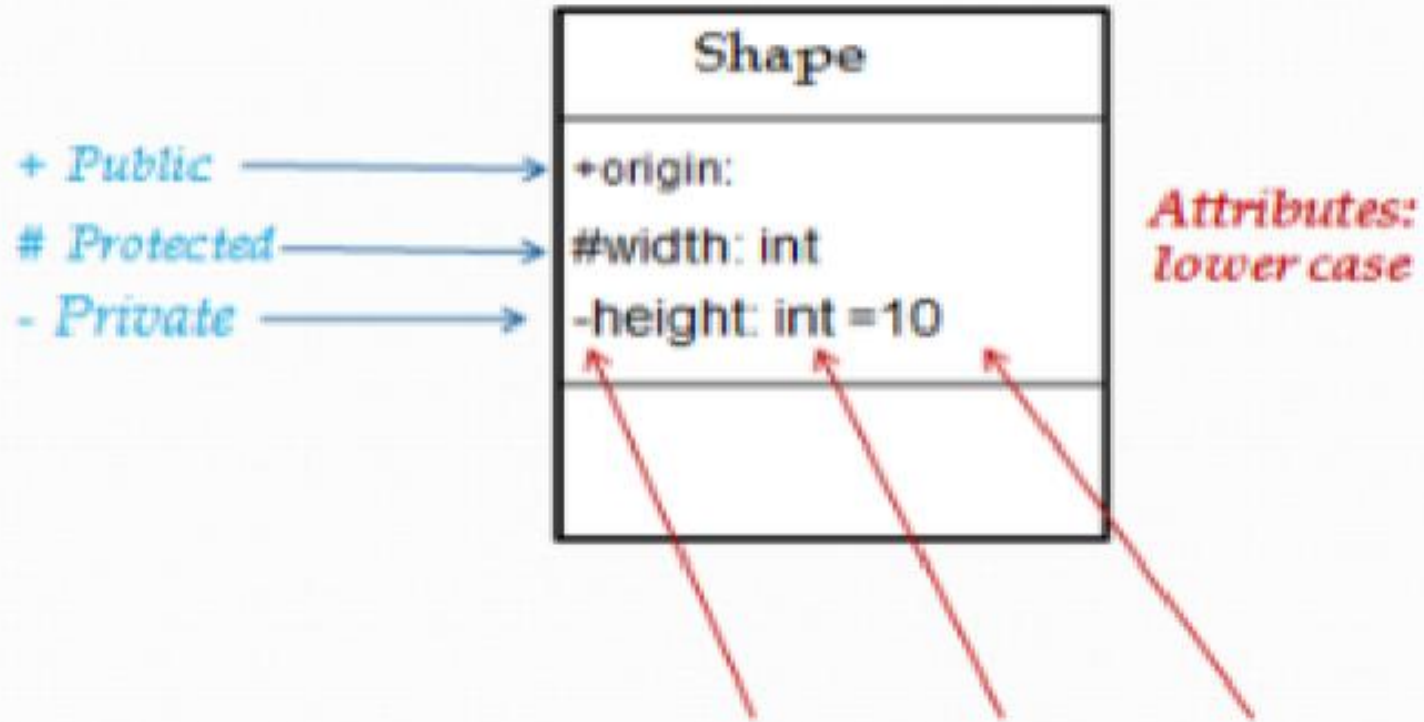
2- **Attributes:** Variables/data members with visibility, type, and optional initial values.

3- **Operations:** Methods/functions with visibility, parameters, and return types.

Attribute and Operation Visibility

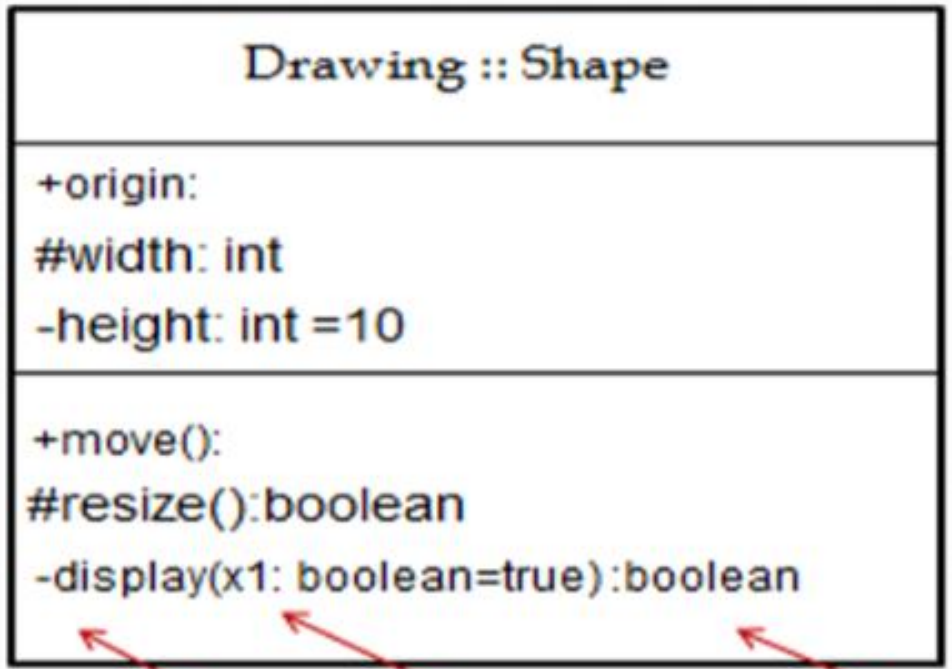
Public (+), Protected (#), Private (-) visibility modifiers.

Class Names And Attributes



Attribute declaration may include visibility, type and initial value

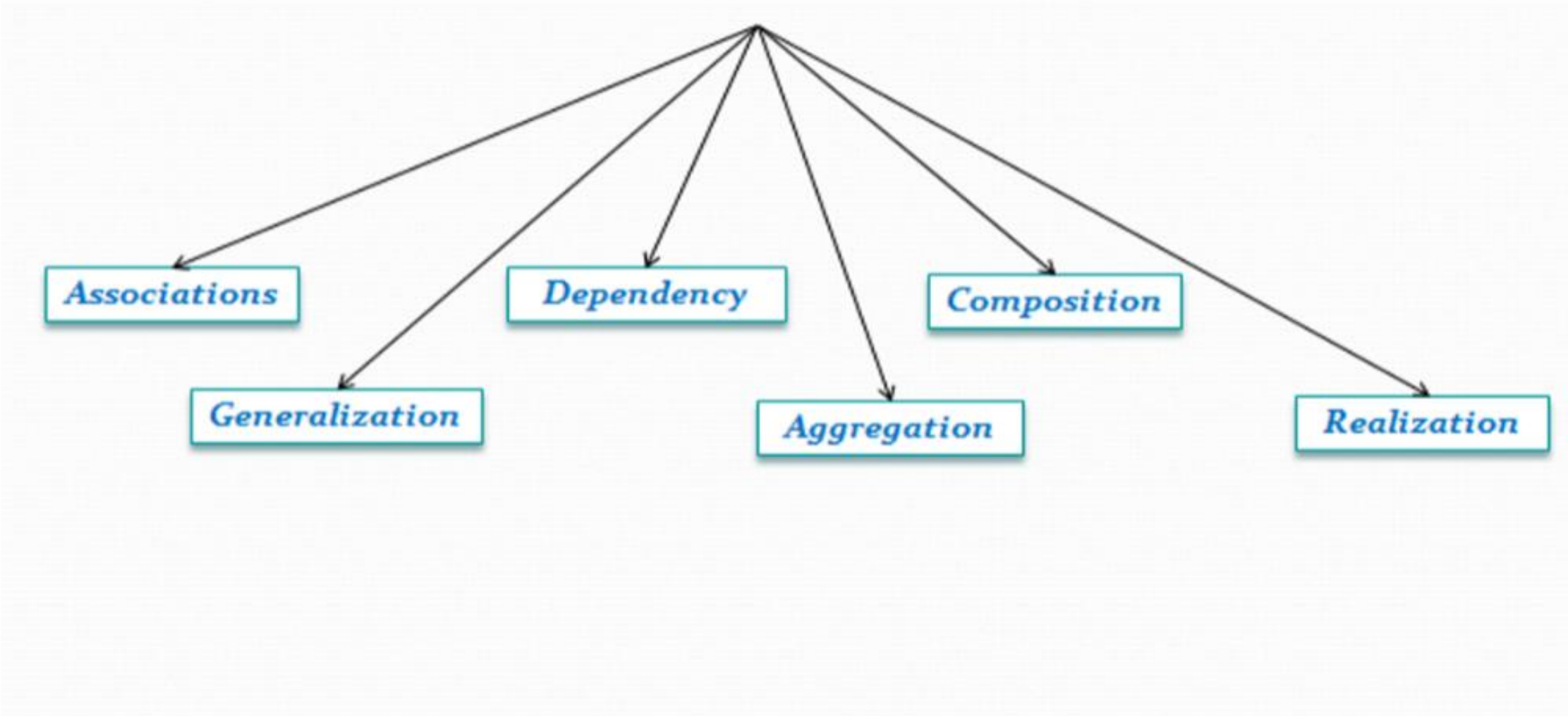
Class Operations



Operations: lower case

- Operation may include visibility, parameters, and return type:
  +opName(param1 : type = initial_value) : return-type

Relationships

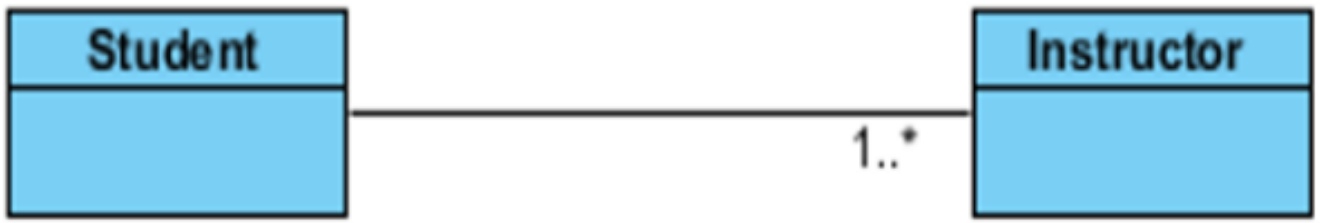


Association and multiplicity

- If two classes in a model need tocommunicate with each other, there must be link between them, and that can berepresented by an association (connector). Association can be represented by a line between these classes with an arrow indicating the navigation direction.

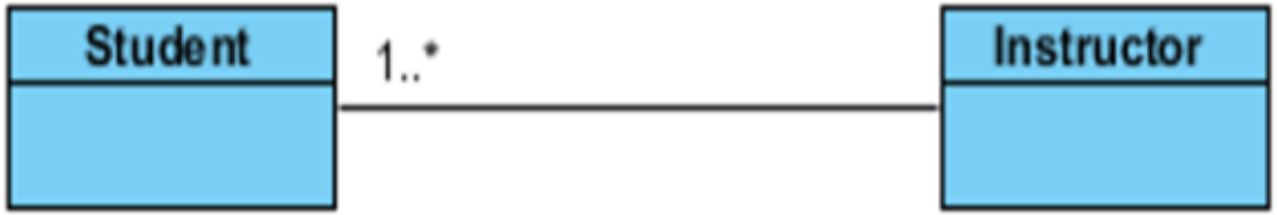
- In case arrow is on the both sides, association has bidirectional association.

- A single student can associate withmultiple teachers:

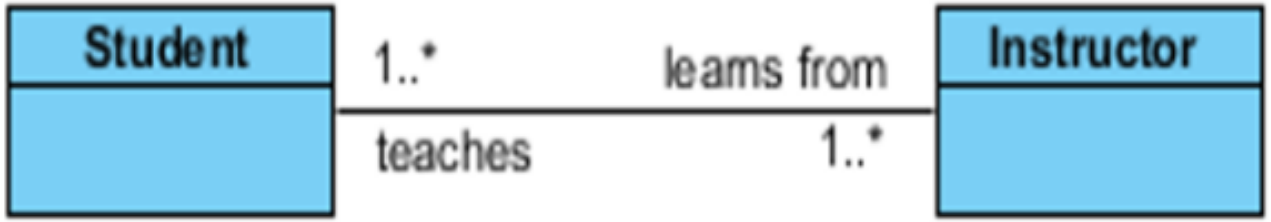


Association and multiplicity

•The example indicates that everyInstructor has one or more Students:



- We can also indicate the behavior of anobject in an association (i.e., the role of an object) using role names.



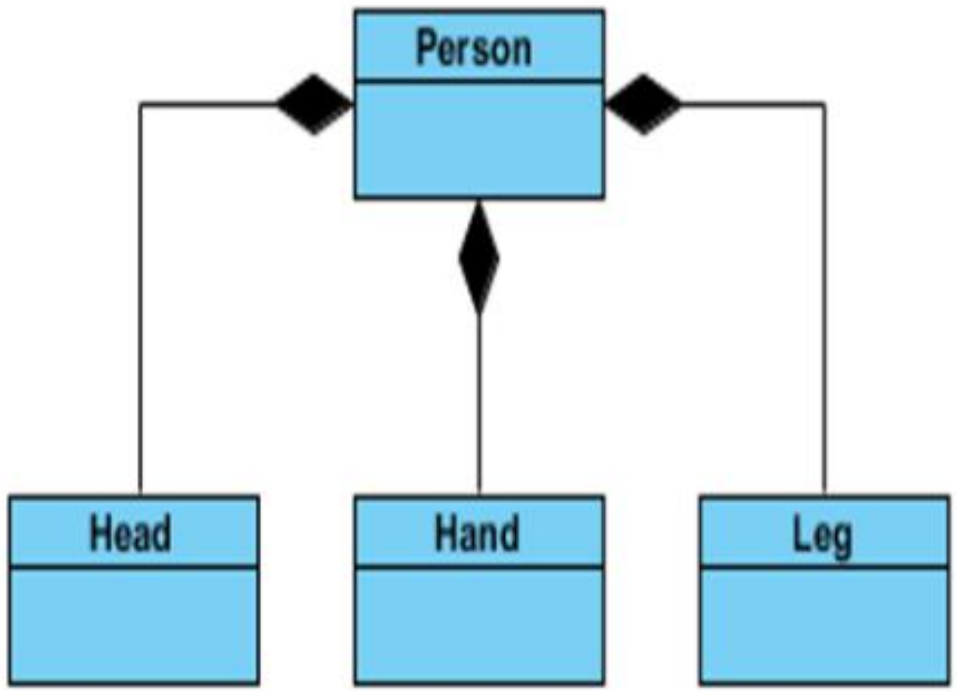
Association vs Aggregation vsComposition

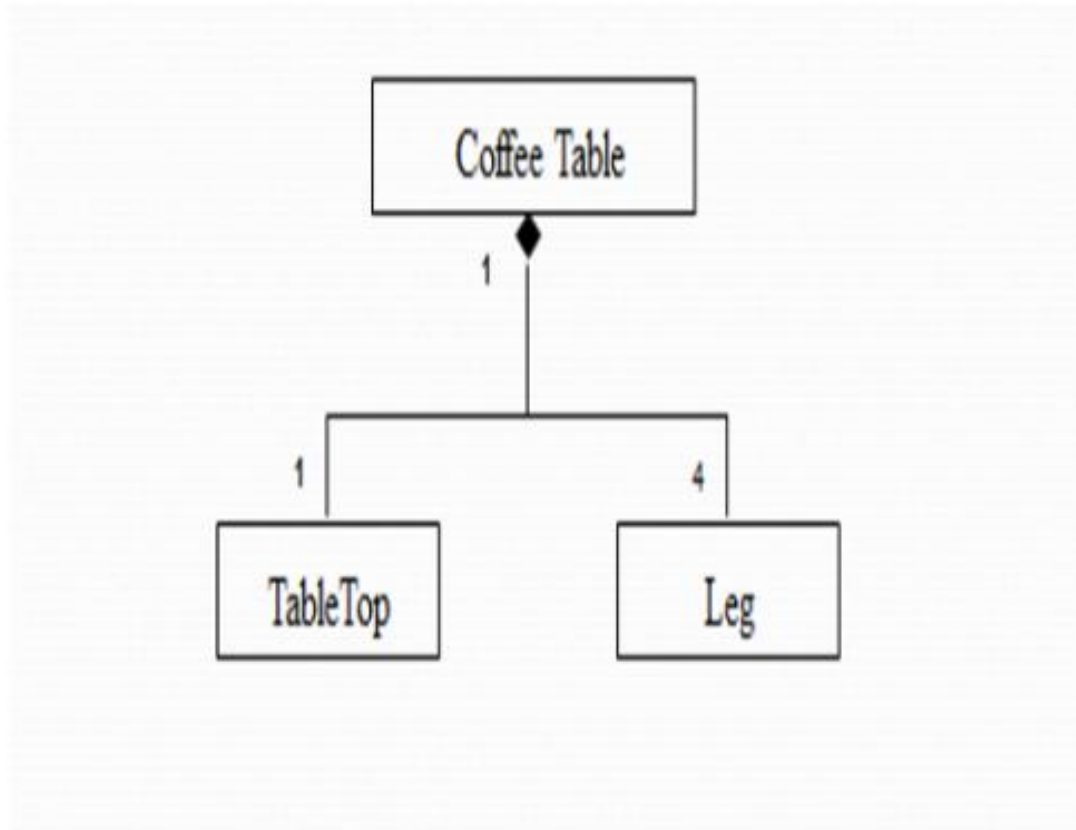
- Actually, **Aggregation and Composition are subsets of association** meaning they are specific cases of association. Inboth aggregation and composition object of one class "owns" object of another class.

- **Aggregation** implies a relationshipwhere the **child can exist independently** of the parent. Example: Class (parent)and Student (child). Delete the Class and the Students still exist.

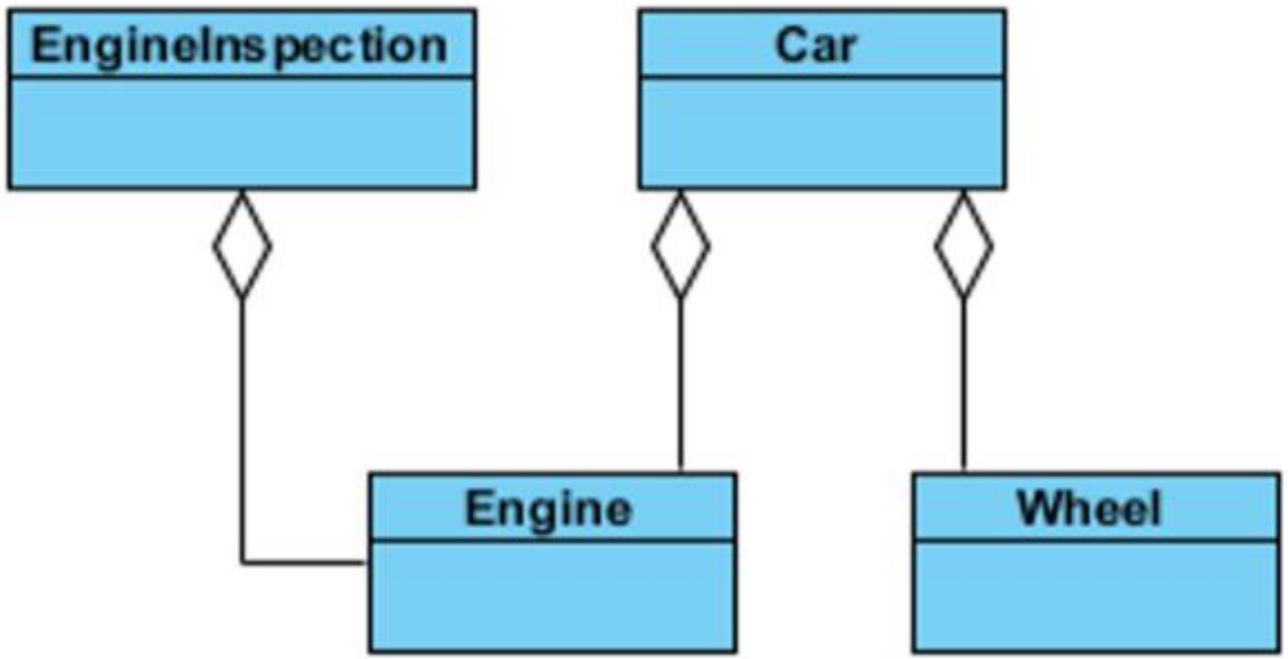
- **Composition** implies a relationshipwhere the **child cannot exist independent** of the parent. Example: House (parent)and Room (child). Rooms don't exist separate to a House.

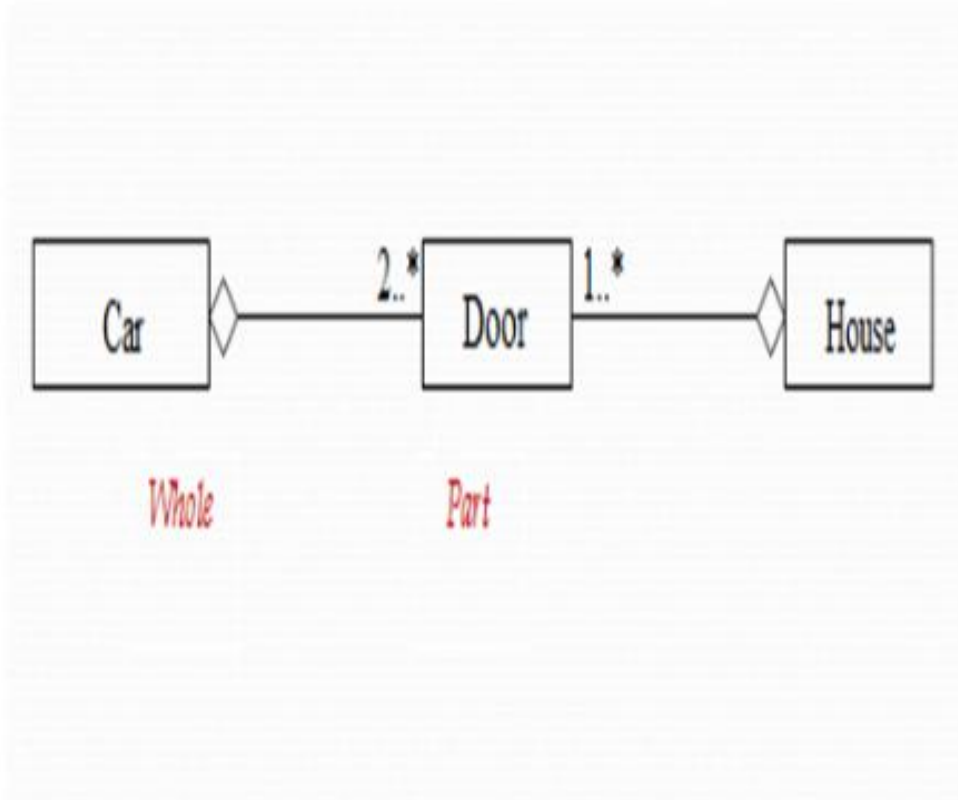
Composition Example





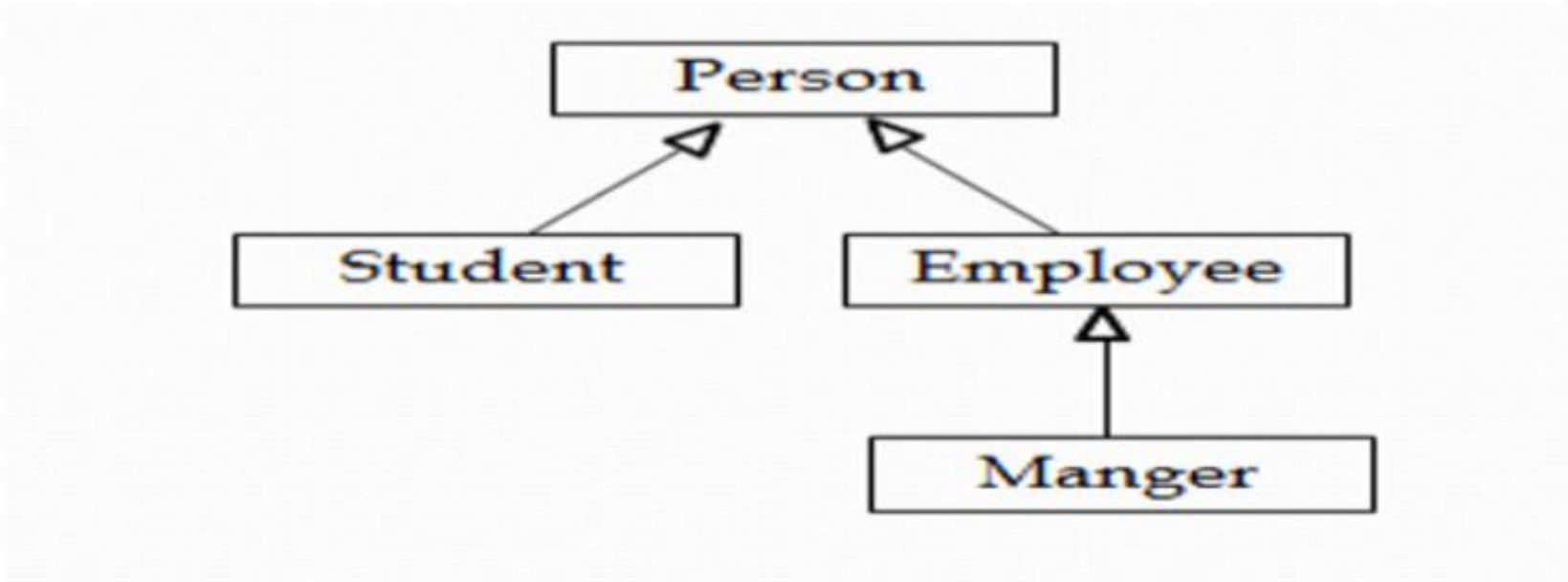
Aggregation Example:





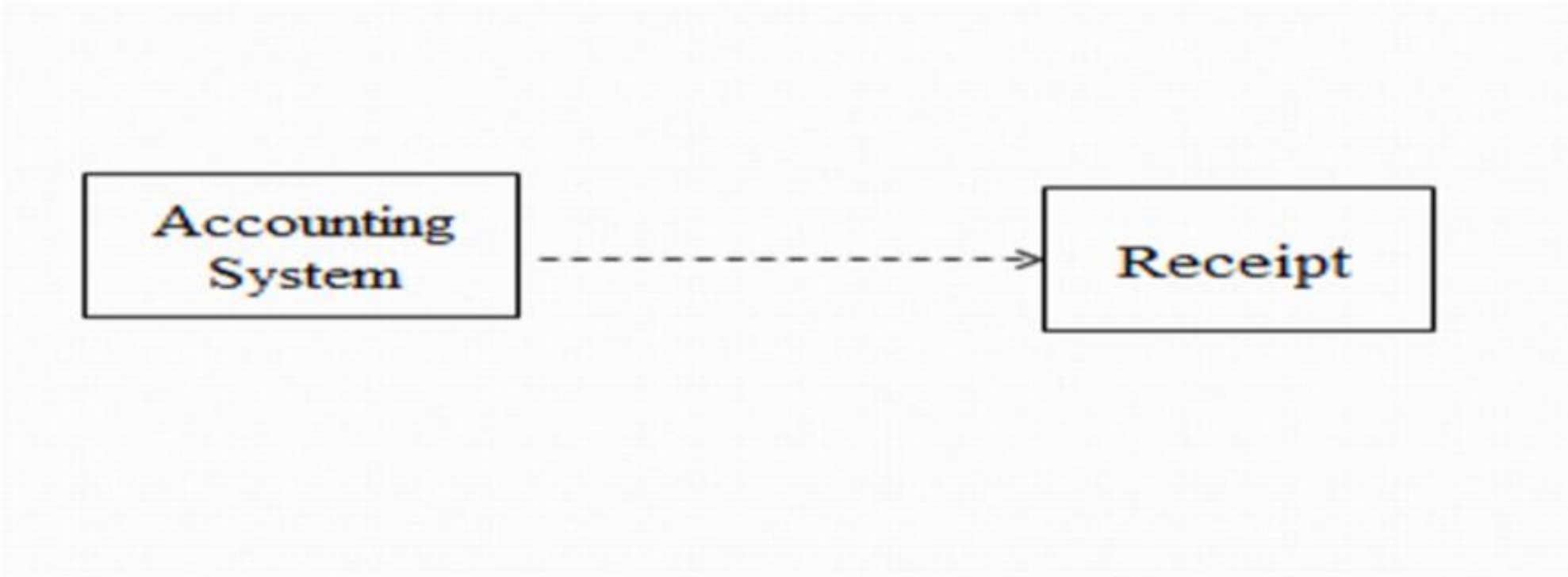
Generalization

- A class might have no parents, inwhich case it's a base class or root class.
- A class might have no children, inwhich case it's a leaf class.
- If a class has exactly one parent,it has single inheritance.
- If a class has more than oneparent, it has multiple inheritance.



Dependencies

- A kind of relationship, one class usesanother. This is called a dependency.

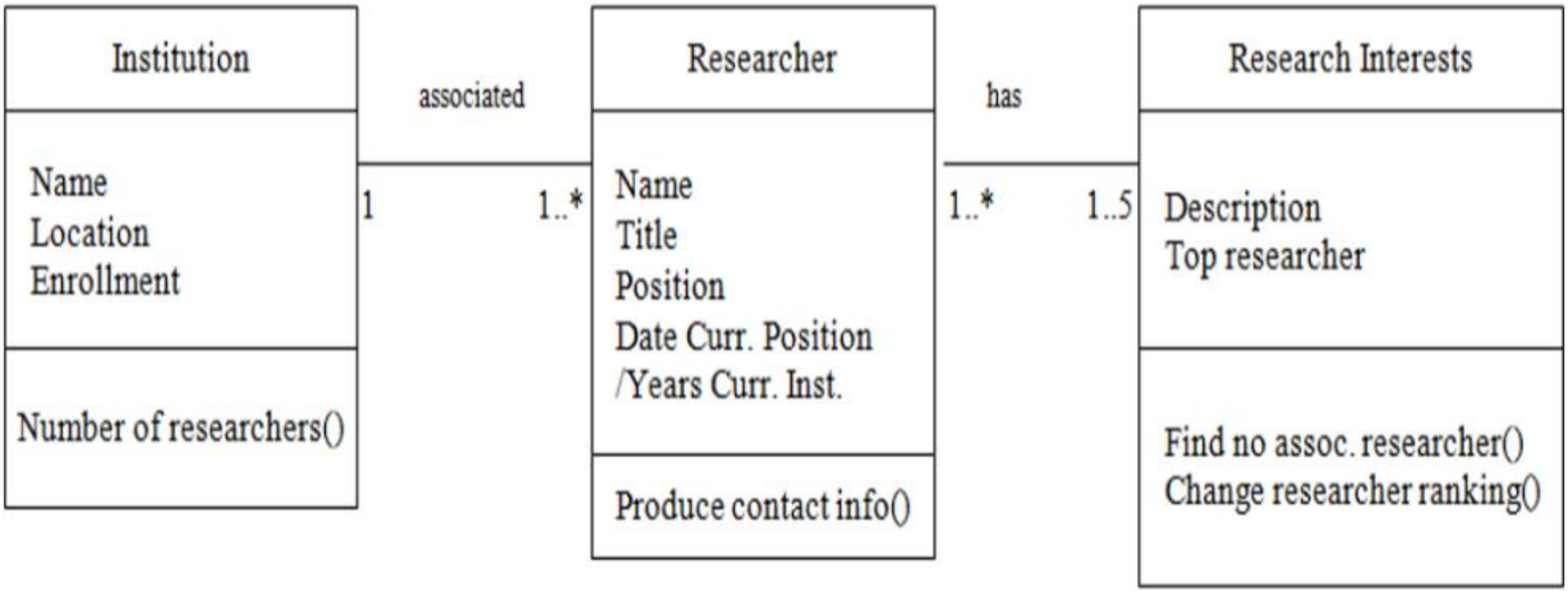



02
Class Diagram Example:
Researcher Database
Technology

Example: Researcher Database

- Createclass diagrams that describe the classes and relationships depicted in thefollowing scenarios:
- Researchers are placed into a database that is maintained by the state ofGeorgia. Information of
- interestincludes researcher name, title, position, date began current position, numberof years
- at current position; university name, location, enrollment; and researchinterests.
- Researchers are associated with one institution, and each researcher can has up to five research interests.
- Morethan one researcher can have the same interest, and the system tracks theranking of the bestresearchersfor each interest.
- Thesystem should be able to insert new researchers, universities, and
- research interests; produce information, such as the number of researchers at eachuniversity, contact information for the researchers, and research interests that do not have associated researchers; and change researcher rankings

Solution


Institution
Name
Location
Enrollment
Number of researchers()
associated
1
1..*
Researcher
Name
Title
Position
Date Curr. Position
/Years Curr. Inst.
Produce contact info()
has
1..*
1..5
Research Interests
Description
Top researcher
Find no assoc. researcher()
Change researcher ranking()

Thanks

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